Ex.No.-2.1-Create a 3D solid model from isometric drawing.

AIM:

To create a 3D model from the given isometric drawing using fusion 360 tool.

REQUIREMENTS:

1. Autodesk fusion 360
2. Windows 10
3. 16 GB of RAM (integrated graphics recommend 1 GB or more)
4. 2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:

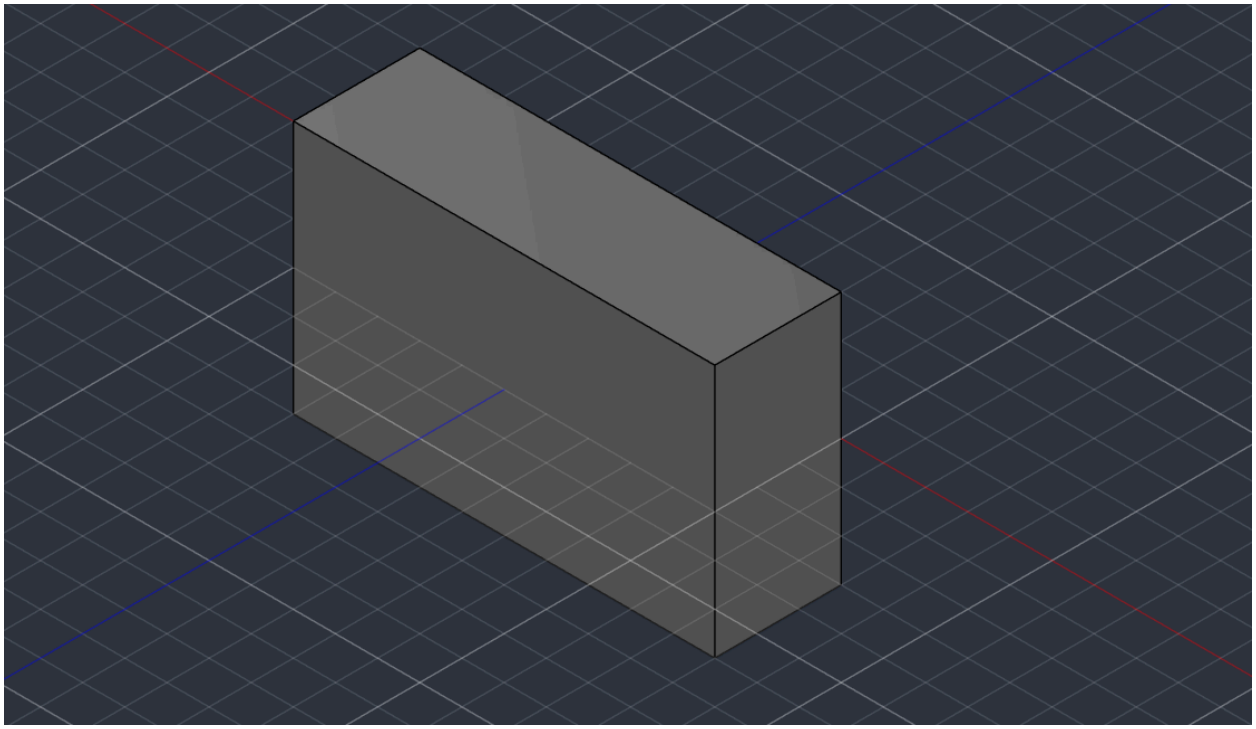
STEP-1:
Start the sketch command
Select Sketch > Create Sketch.

STEP-2:
Select the Sketch Plane
Select the Front (XZ) Plane

STEP-3:
Create a Rectangle
Select Sketch > Rectangle > Center Rectangle.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the rectangle.
Type the Tab button and enter the values 100 mm and 60mm for the width and height respectively.
Click again to place the rectangle.
Click Finish Sketch after completing.

STEP-4:
Extrude Rectangle in XZ Plane
Select Extrude.
Select the Rectangle sketch.
Click again to place the value.
Type in the value of 30mm.

Press ENTER to accept the value.



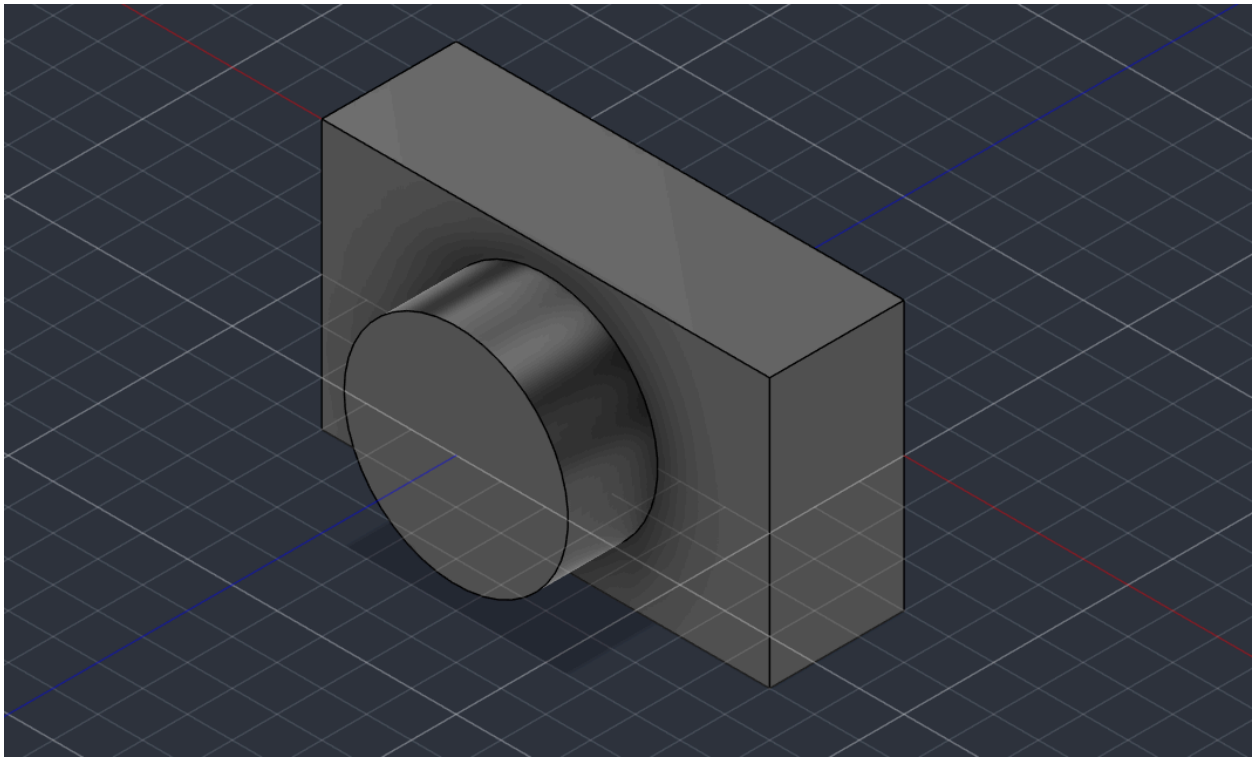
STEP-5:
Start the sketch command
Select Sketch > Create Sketch.

STEP-6:
Select the Sketch Plane
Select the FRONT SURFACE of rectangular block

STEP-7:
Create a CIRCLE
Select Sketch >CIRCLE > CENTER CIRCLE.
Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.
Click once to begin placing the CIRCLE.
Type the Tab button and enter the value DIA 50mm .
Click again to place the circle.
Click Finish Sketch after completing.

STEP-8:
Extrude CIRCLE
Select Extrude.
Select the circle sketch.
Click again to place the value.
Type in the value of 20mm.

Press ENTER to accept the value.

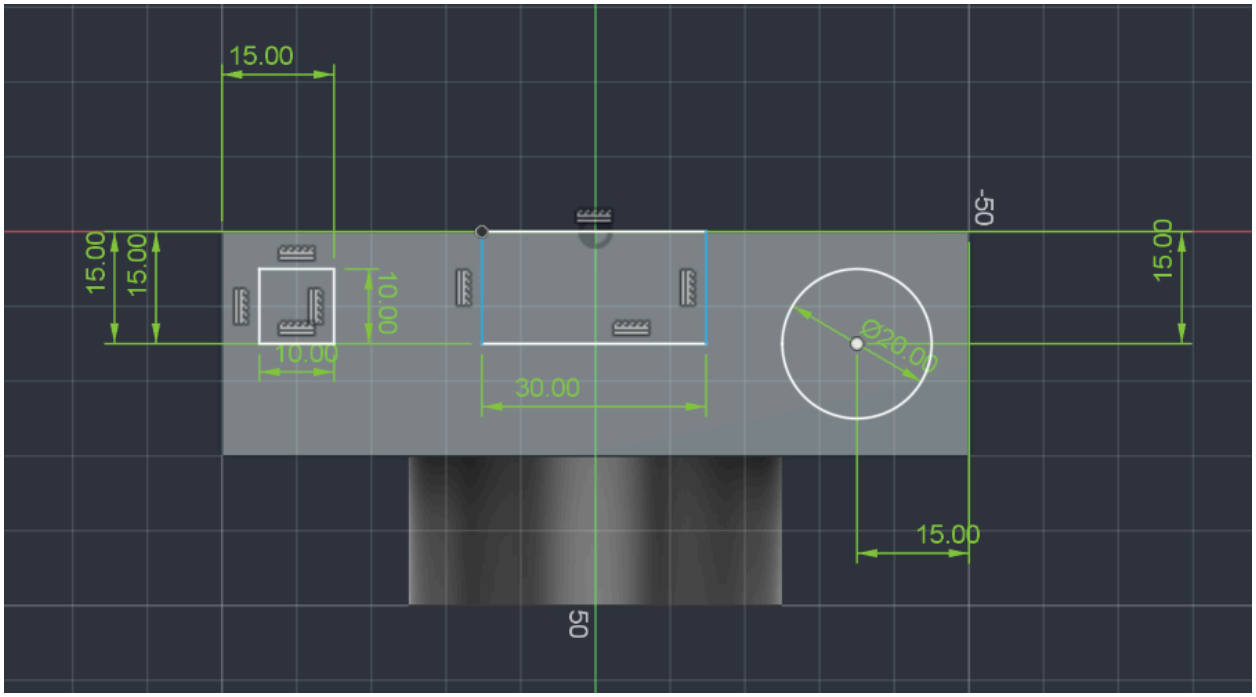


STEP-9:
Start the sketch command
Select Sketch > Create Sketch.

STEP-10:
Select the Sketch Plane
Select the TOP SURFACE OF RECTANGLE.

STEP-11:
Draw a circle of diameter 20mm on the right side.
Fix its dimensions so that its centre and both sides of the rectangle are 15mm apart.
On the left, draw a square of side 10mm.
Fix its dimension such that its is 5mm apart from top left corner of the rectangle.

Draw a rectangle of 30mm length and 15 mm breadth at the top-centre.



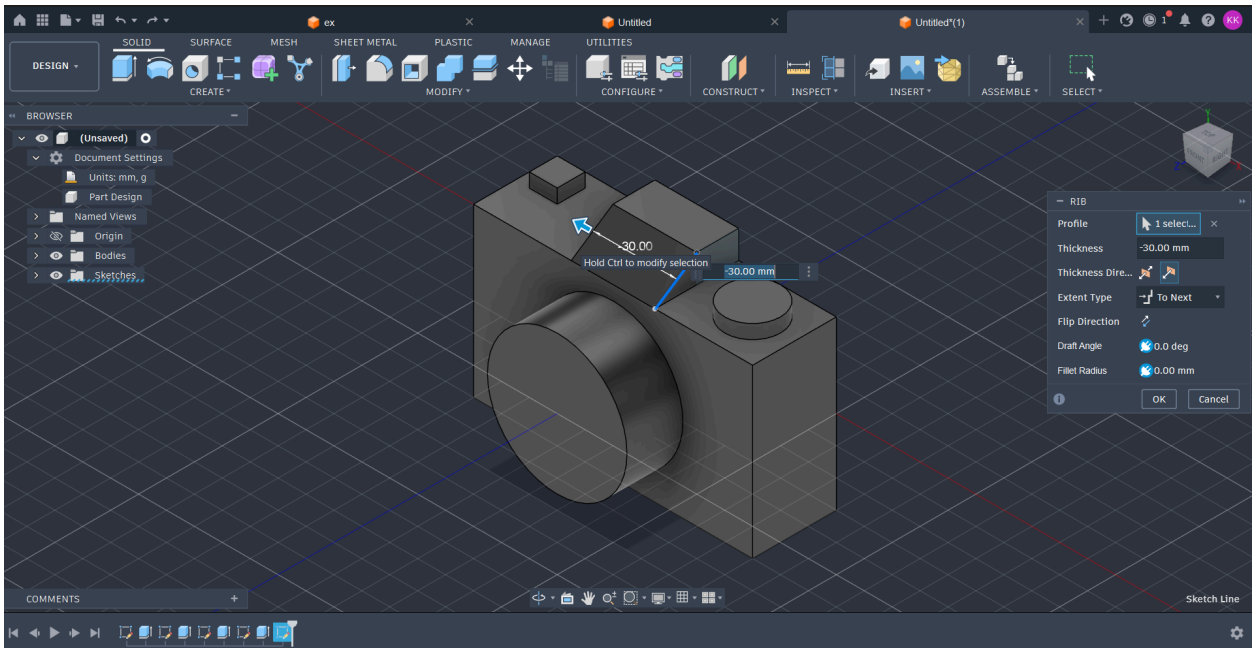
STEP-12:
Extrude the circle and the square.
Select extrude.
Select both sketchs.
Click again to place the value.
Type in the value of 5mm.
Press ENTER to accept the value.

STEP-13:
Extrude the rectangle.
Select extrude.
Select the rectangle sketch.
Click again to place the value.
Type in the value of 10mm.
Press ENTER to accept the value

STEP-14:
Start the sketch command
Select Sketch > Create Sketch

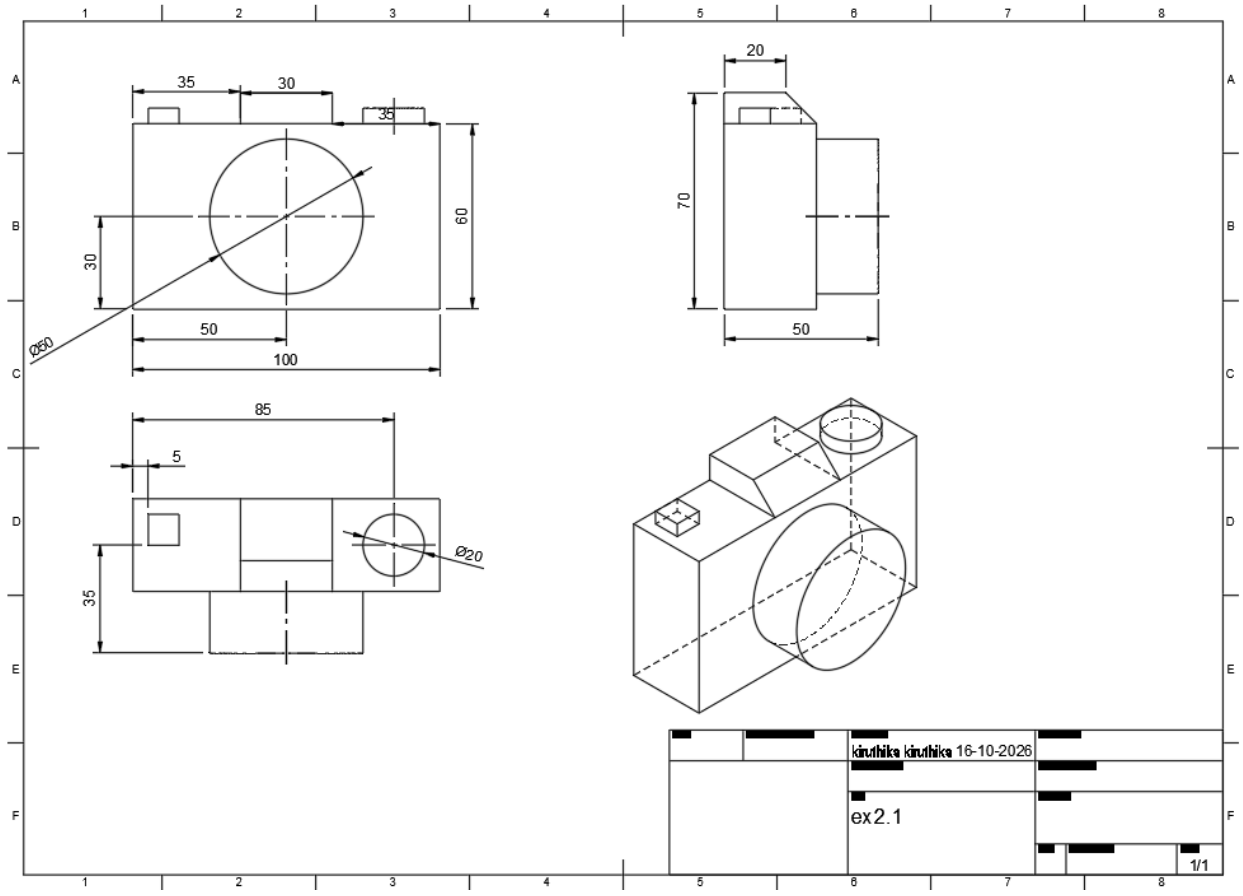
STEP-15:
Draw a line connecting the smaller and bigger rectangle.
Click Finish Sketch after completing.

STEP-16:
Select create > Create Rib.
Select the line.
Type in the value of 30mm.
Press ENTER to accept the value.



STEP-17:
Switch to drawing mode to plot the sketch in ISO standard.

OUTPUT:



RESULT:

Thus, the 3D model is created from given isometric drawing.